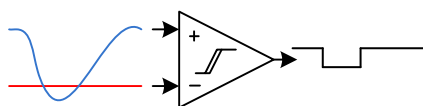
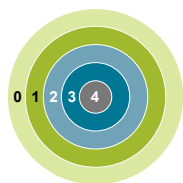


## 29. ACMP - Analog Comparator



### Quick Facts

#### What?

The ACMP (Analog Comparator) compares two analog signals and returns a digital value telling which is greater.

#### Why?

Applications often do not need to know the exact value of an analog signal, only if it has passed a certain threshold. Often the voltage must be monitored continuously, which requires extremely low power consumption.

#### How?

Available down to Energy Mode 3, the ACMP can wake up the system when input signals pass the threshold. The analog comparator can compare two analog signals or one analog signal and a highly configurable internal reference.

### 29.1 Introduction

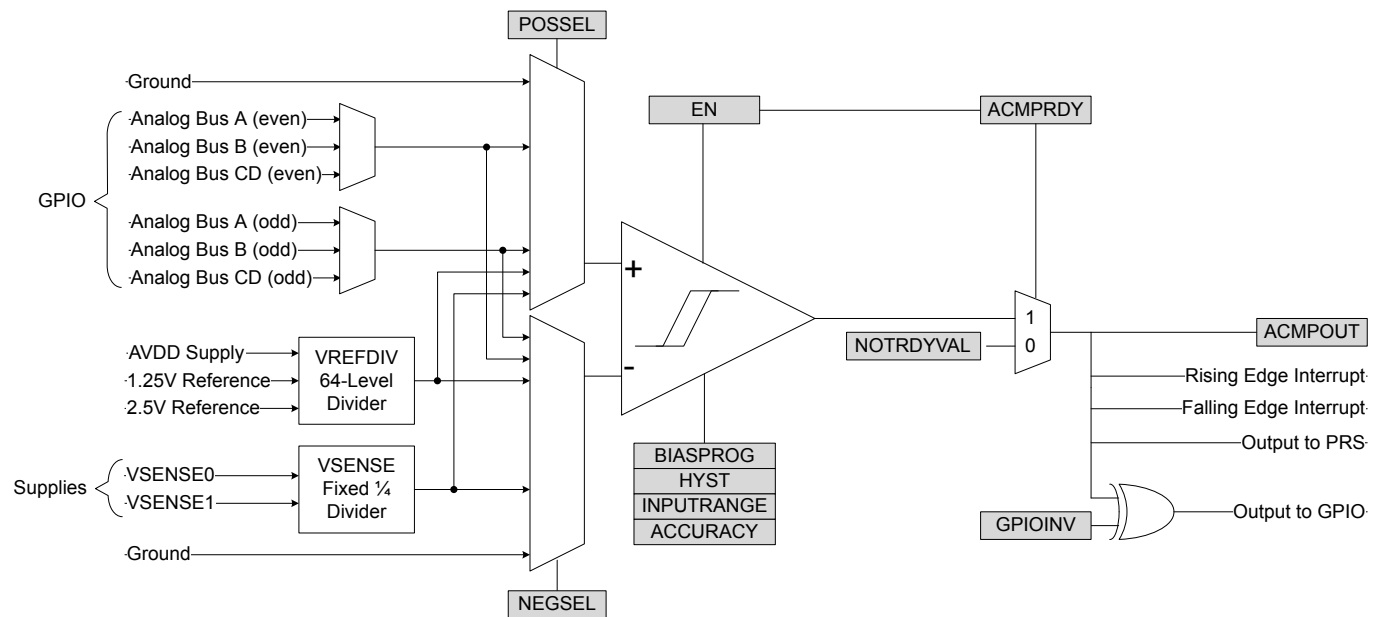
The Analog Comparator compares the voltage of two analog inputs and outputs a digital signal indicating which input voltage is higher. Inputs can either be from internal references or from external pins. Response time, and thereby the current consumption, can be configured by altering the current supply to the comparator.

### 29.2 Features

- Internal and external input selections:
  - External port I/O routed via ABUS
  - Internal 1.25 V bandgap reference voltage with programmable divider
  - Internal 2.5 V bandgap reference voltage with programmable divider
  - AVDD supply voltage with programmable divider
  - VDAC auxiliary outputs
- Voltage supply monitoring
- Configurable hysteresis
- Selectable response time
- Operational in EM0 to EM3
- Asynchronous interrupt generation on selectable edges
- Configurable output state when inactive
- Output routing options:
  - Route to GPIO via DBUS
  - Route to most peripherals via PRS

### 29.3 Functional Description

An overview of the ACMP is shown in [Figure 29.1 ACMP Overview on page 1110](#).



**Figure 29.1. ACMP Overview**

The comparator has two analog inputs: one positive and one negative. When the comparator is active, the output indicates which of the two input voltages is higher. When the voltage on the positive input is higher than the voltage on the negative input, the digital output is high and vice versa.

In addition to the comparator core, the ACMP front-end includes reference sources, voltage divider circuits, and input muxes to route signals to the positive and negative inputs. The output from the ACMP is available on both PRS and GPIO, in addition to being observable in the ACMP\_STATUS register.

### 29.3.1 Configuration and Control

The ACMP is configured and controlled through three registers: ACMP\_CFG, ACMP\_CTRL, and ACMP\_INPUTCTRL. Configuration through ACMP\_CFG needs to happen before the ACMP is enabled. The control registers ACMP\_CTRL and ACMP\_INPUTCTRL can only be updated after the ACMP is enabled. The ACMP is enabled by setting the EN bit in ACMP\_EN. If ACMP\_CFG is updated when EN = 1, or ACMP\_CTRL / ACMP\_INPUTCTRL is updated while EN = 0, a bus fault is issued.

The input muxes are configured in the POSSEL / NEGSEL bitfields in ACMP\_INPUTCTRL. All references and inputs are available in the modes defined for these two registers. The INPUTCTRL bit in ACMP\_SYNCBUSY should be checked before writing to ACMP\_INPUTCTRL. If the ACMP\_SYNCBUSY\_INPUTCTRL bit is 1, it means a previous write to the ACMP\_INPUTCTRL register is pending, and software should wait until ACMP\_SYNCBUSY\_INPUTCTRL bit reads 0.

The POSSEL and NEGSEL muxes share several resources on the device, such as the VREFDIV and VSENSE divider circuits. Thus, there are some constraints on the POSSEL / NEGSEL configurations:

- POSSEL and NEGSEL cannot select an even numbered GPIO pin at the same time.
- POSSEL and NEGSEL cannot select an odd numbered GPIO pin at the same time.
- POSSEL and NEGSEL cannot both select a supply voltage via one of the VSENSE inputs.
- POSSEL and NEGSEL cannot both select an input using VREFDIV.
- If POSSEL = EXTPx, a low power reference (postfixed with LP) cannot be selected for NEGSEL.

If one of these constraints are violated, the INPUTCONFLICT status flag and INPUTCONFLICTIF interrupt flag will be set.

The ACMP also uses shared chip-level analog bus resources to connect to external GPIO pins. Which bus the ACMP is using depends on the configuration of POSSEL and NEGSEL. To allow the ACMP to control an analog bus, the bus must be allocated to ACMP in the GPIO module, using the GPIO\_xBUSALLOC registers. For example, pin PB5 is an odd-numbered pin on port PB, and could connect via either analog bus BODD0 or BODD1. This is configured using the BODD0 or BODD1 field in GPIO\_BBUSALLOC.

If the ACMP peripheral is trying to access a bus that has not been allocated to that instance of ACMP, the PORTALLOCERR status flag and PORTALLOCERRIF interrupt flag will be set.

### 29.3.2 Warmup Time

When the comparator is enabled or the input muxes are reconfigured, it requires some time to stabilize. On first enable (ACMP\_EN\_EN = 1), the comparator core requires 2.5 us to stabilize. In addition to this, any references selected may require some time to warm up. See [Table 29.1 Warmup Time on page 1111](#) for warmup times for the different references. When reconfiguring the ACMP (without disabling it), only the warmup times given in the table will be observed. When the comparator is ready for use, the ACMPRDY status flag and the ACMPRDY interrupt flag will be set.

**Note:** The hardware timeout is not sufficient to ensure glitch-free operation when using BIAS<=3. To avoid startup glitches, software should wait for 60 us after enabling the comparator to use the output.

During the warmup time and when the comparator is inactive, the comparator output will be set to the state defined by the NOTRDYV-AL bit in ACMP\_CTRL.

**Table 29.1. Warmup Time**

| Reference                                  | Warmup time |
|--------------------------------------------|-------------|
| Low power reference: POSSEL / NEGSEL = *LP | 10 us       |
| VSENSE: POSSEL / NEGSEL = VSENSE*          | 5 us        |
| VREF: POSSEL / NEGSEL = VREF*              | 2 us        |
| None of the above                          | 0.5 us      |

### 29.3.3 Response Time

There is a delay from when the input voltage changes polarity to when the output toggles. This delay is called the response time and can be altered by increasing or decreasing the bias current to the comparator through the BIASPROG bitfield in the ACMP\_CFG register. The current and speed of the circuit increase as the value of BIASPROG is increased. See the part datasheet for specific current and response times related to setting of BIASPROG.

### 29.3.4 Hysteresis

When the hysteresis level is set to a non-zero value, the digital output will not toggle until the positive input voltage is at a voltage equal to the hysteresis level above or below the negative input voltage (see [Figure 29.2 Hysteresis on page 1112](#)). This feature can be used to avoid continual comparator output changes due to input noise when the positive and negative inputs are similar. Hysteresis requires the input difference to exceed the hysteresis threshold before the output can change and can reject limited amounts of noise. The hysteresis in ACMP can be configured to three different levels (10 mV, 20 mV, 30 mV), and can be enabled on positive (rising), negative (falling), or both edges. Hysteresis is configured in the HYST bitfield in ACMP\_CFG.

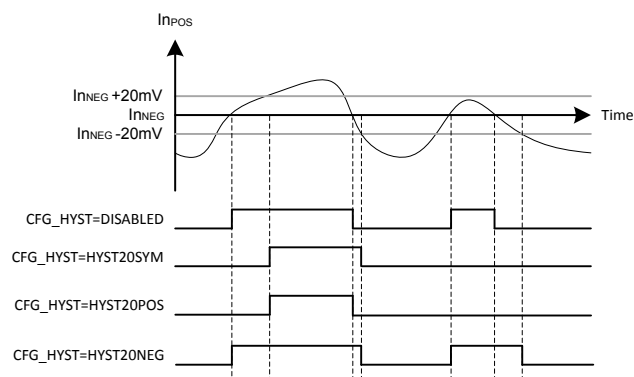


Figure 29.2. Hysteresis

### 29.3.5 Supply Voltage Monitoring (VSENSE)

The ACMP can be used to monitor supply voltages. This is done by selecting VSENSE01DIV4(LP) or VSENSE11DIV4(LP) for either POSSEL or NEGSEL. Note that the input to the comparator core will be divided by 4, as illustrated in [Figure 29.1 ACMP Overview on page 1110](#). To reduce energy consumption, a sample/hold circuit can be used to periodically sample the power supplies. To enable this, select VSENSE01DIV4LP or VSENSE11DIV4LP in POSSEL / NEGSEL. Because the sample/hold feature uses the comparator in a non-continuous fashion, enabling this will increase response times and reduce the accuracy of the comparator. The connections between VSENSE0 and VSENSE1 to power supplies are summarized in the table below.

Table 29.2. VSENSE connections

| ACMP instance | VSENSE0 | VSENSE1       |
|---------------|---------|---------------|
| ACMP0         | AVDD    | VDDIO0        |
| ACMP1         | DVDD    | Not connected |

### 29.3.6 VREFDIV Sources

The ACMP has two internal bandgap references: 2.5 V and 1.25 V. In addition, AVDD can be used as a reference. To select one of these references, configure POSSEL / NEGSEL to VREFDIVAVDD, VREFDIV1V25, or VREFDIV2V5. The ACMP also includes sample/hold functionality to reduce energy consumption. To enable the sample/hold feature, select VREFDIVAVDDL, VREFDIV1V25LP, or VREFDIV2V5LP. These references can be divided by configuring VREFDIV in ACMP\_INPUTCTRL. This division factor will be VREFDIV / 63, such that  $VREFOUT = VREFIN * (VREFDIV / 63)$ .

### 29.3.7 Input Range and Accuracy Settings

By default, the ACMP can accept external rail-to-rail inputs, from 0 to AVDD. If external voltages will never be higher than AVDD - 0.7 V, the INPUTRANGE bit in ACMP\_CFG can be set to 1 to reduce the power consumption of the block.

The ACMP also has an adjustable accuracy setting (ACCURACY in ACMP\_CFG). ACCURACY is set to LOW by default, which conserves power, but may have degraded performance for rapidly changing analog Port selections in either the ACMP or the GPIO. ACCURACY can be set to HIGH to insure ACMP accuracy (at the expense of extra power consumption), when configuration changes are expected at a high rate (more than once per ms, for example), such as when scanning through channels.

### 29.3.8 Interrupts and PRS Output

The analog comparator includes independent output flags for rising edge (RISEIF) and falling edge (FALLIF) events. These will be set when a rising or falling edge is detected, respectively.

Three other interrupt sources are also available. PORTALLOCCERRIF and INPUTCONFLICTIF are input configuration error flags, detailed in [29.3.1 Configuration and Control](#). The ACMPRDYIF flag indicates comparator stability after the warmup period.

The comparator output is available as an asynchronous PRS producer, and can be routed to other peripherals in the system via PRS.

### 29.3.9 Output to GPIO

The output from the comparator is available as alternate functions to the GPIO pins. Each pin connection can be enabled/disabled separately using the GPIO module control registers. See the device data sheet for the available locations for each signal.

The GPIO pin must also be set as output. The output to the GPIO can be inverted by setting the GPIOINV bit in ACMP\_CTRL.

## 29.4 ACMP Register Map

The offset register address is relative to the registers base address.

| Offset | Name                               | Type        | Description               |
|--------|------------------------------------|-------------|---------------------------|
| 0x000  | <a href="#">ACMP_IPVERSION</a>     | R           | IP Version ID             |
| 0x004  | <a href="#">ACMP_EN</a>            | RW ENABLE   | ACMP Enable               |
| 0x008  | <a href="#">ACMP_SWRST</a>         | RW SWRST    | Software Reset            |
| 0x00C  | <a href="#">ACMP_CFG</a>           | RW CONFIG   | Configuration Register    |
| 0x010  | <a href="#">ACMP_CTRL</a>          | RW          | Control Register          |
| 0x014  | <a href="#">ACMP_INPUTCTRL</a>     | RW SYNC     | Input Control Register    |
| 0x018  | <a href="#">ACMP_STATUS</a>        | RH          | Status Register           |
| 0x01C  | <a href="#">ACMP_IF</a>            | RWH INTFLAG | Interrupt Flag Register   |
| 0x020  | <a href="#">ACMP_IEN</a>           | RW          | Interrupt Enable Register |
| 0x024  | <a href="#">ACMP_SYNCBUSY</a>      | RH          | Syncbusy                  |
| 0x1000 | <a href="#">ACMP_IPVERSION_SET</a> | R           | IP Version ID             |
| 0x1004 | <a href="#">ACMP_EN_SET</a>        | RW ENABLE   | ACMP Enable               |
| 0x1008 | <a href="#">ACMP_SWRST_SET</a>     | RW SWRST    | Software Reset            |
| 0x100C | <a href="#">ACMP_CFG_SET</a>       | RW CONFIG   | Configuration Register    |
| 0x1010 | <a href="#">ACMP_CTRL_SET</a>      | RW          | Control Register          |
| 0x1014 | <a href="#">ACMP_INPUTCTRL_SET</a> | RW SYNC     | Input Control Register    |
| 0x1018 | <a href="#">ACMP_STATUS_SET</a>    | RH          | Status Register           |
| 0x101C | <a href="#">ACMP_IF_SET</a>        | RWH INTFLAG | Interrupt Flag Register   |
| 0x1020 | <a href="#">ACMP_IEN_SET</a>       | RW          | Interrupt Enable Register |
| 0x1024 | <a href="#">ACMP_SYNCBUSY_SET</a>  | RH          | Syncbusy                  |
| 0x2000 | <a href="#">ACMP_IPVERSION_CLR</a> | R           | IP Version ID             |
| 0x2004 | <a href="#">ACMP_EN_CLR</a>        | RW ENABLE   | ACMP Enable               |
| 0x2008 | <a href="#">ACMP_SWRST_CLR</a>     | RW SWRST    | Software Reset            |
| 0x200C | <a href="#">ACMP_CFG_CLR</a>       | RW CONFIG   | Configuration Register    |
| 0x2010 | <a href="#">ACMP_CTRL_CLR</a>      | RW          | Control Register          |
| 0x2014 | <a href="#">ACMP_INPUTCTRL_CLR</a> | RW SYNC     | Input Control Register    |
| 0x2018 | <a href="#">ACMP_STATUS_CLR</a>    | RH          | Status Register           |
| 0x201C | <a href="#">ACMP_IF_CLR</a>        | RWH INTFLAG | Interrupt Flag Register   |
| 0x2020 | <a href="#">ACMP_IEN_CLR</a>       | RW          | Interrupt Enable Register |
| 0x2024 | <a href="#">ACMP_SYNCBUSY_CLR</a>  | RH          | Syncbusy                  |
| 0x3000 | <a href="#">ACMP_IPVERSION_TGL</a> | R           | IP Version ID             |
| 0x3004 | <a href="#">ACMP_EN_TGL</a>        | RW ENABLE   | ACMP Enable               |
| 0x3008 | <a href="#">ACMP_SWRST_TGL</a>     | RW SWRST    | Software Reset            |
| 0x300C | <a href="#">ACMP_CFG_TGL</a>       | RW CONFIG   | Configuration Register    |
| 0x3010 | <a href="#">ACMP_CTRL_TGL</a>      | RW          | Control Register          |

| Offset | Name                               | Type        | Description               |
|--------|------------------------------------|-------------|---------------------------|
| 0x3014 | <a href="#">ACMP_INPUTCTRL_TGL</a> | RW SYNC     | Input Control Register    |
| 0x3018 | <a href="#">ACMP_STATUS_TGL</a>    | RH          | Status Register           |
| 0x301C | <a href="#">ACMP_IF_TGL</a>        | RWH INTFLAG | Interrupt Flag Register   |
| 0x3020 | <a href="#">ACMP_IEN_TGL</a>       | RW          | Interrupt Enable Register |
| 0x3024 | <a href="#">ACMP_SYNCBUSY_TGL</a>  | RH          | Syncbusy                  |

## 29.5 ACMP Register Description

### 29.5.1 ACMP\_IPVERSION - IP Version ID

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|
| 0x000  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reset  | 0x2          |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Access | R            |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |
| Name   | IPVERSION    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |

| Bit  | Name      | Reset | Access | Description          |
|------|-----------|-------|--------|----------------------|
| 31:0 | IPVERSION | 0x2   | R      | <b>IP version ID</b> |

## 29.5.2 ACMP\_EN - ACMP Enable

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |           |     |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|-----------|-----|
| 0x004  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |           |     |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | 0x0       | 0x0 |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | R         | RW  |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | DISABLING | EN  |

| Bit  | Name      | Reset                                                                                                                                                                      | Access | Description                                                                                                                                                                                                                                            |
|------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:2 | Reserved  | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                                                                                                                                                                                                                                        |
| 1    | DISABLING | 0x0                                                                                                                                                                        | R      | <b>Disablement busy status</b><br><br>When En is cleared, DISABLING is set immediately, and cleared when disablement finishes. Disablement resets peripheral cores and not APB registers except hardware updated registers such as INTFLAGS and FIFOs. |
| 0    | EN        | 0x0                                                                                                                                                                        | RW     | <b>Module enable</b><br><br>The ENABLE bit enables the module. Software should write to CONFIG type registers before setting the ENABLE bit. Software should write to SYNC type registers only after setting the ENABLE bit.                           |

## 29.5.3 ACMP\_SWRST - Software Reset

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |           |       |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|-----------|-------|
| 0x008  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |           |       |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | 0x0       | 0x0   |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | R         | W     |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | RESETTING | SWRST |

| Bit  | Name      | Reset                                                                                                                                                                      | Access | Description                       |
|------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------|
| 31:2 | Reserved  | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                   |
| 1    | RESETTING | 0x0                                                                                                                                                                        | R      | <b>Software reset busy status</b> |
| 0    | SWRST     | 0x0                                                                                                                                                                        | W      | <b>Software reset</b>             |



## 29.5.4 ACMP\_CFG - Configuration Register

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |          |            |    |    |    |    |      |   |   |   |   |   |   |   |      |   |   |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----------|------------|----|----|----|----|------|---|---|---|---|---|---|---|------|---|---|
| 0x00C  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16       | 15         | 14 | 13 | 12 | 11 | 10   | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2    | 1 | 0 |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 0x0      | 0x0        |    |    |    |    | 0x0  |   |   |   |   |   |   |   | 0x4  |   |   |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    | RW       | RW         |    |    |    |    | RW   |   |   |   |   |   |   |   | RW   |   |   |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    | ACCURACY | INPUTRANGE |    |    |    |    | HYST |   |   |   |   |   |   |   | BIAS |   |   |

| Bit   | Name       | Reset                                                                                                                                                                      | Access                                                                                                          | Description                                                                                                                                                                                                                                                                              |
|-------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:18 | Reserved   | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |                                                                                                                 |                                                                                                                                                                                                                                                                                          |
| 17    | ACCURACY   | 0x0                                                                                                                                                                        | RW                                                                                                              | <b>ACMP accuracy mode</b><br><br>Setting ACCURACY to HIGH reduces the noise in the signal input path of the ACMP. Note, high frequency changes can cause the ACMP performance to degrade. For such uses, such as quickly scanning through multiple channels, this should be set to HIGH. |
|       | Value      | Mode                                                                                                                                                                       | Description                                                                                                     |                                                                                                                                                                                                                                                                                          |
|       | 0          | LOW                                                                                                                                                                        | ACMP operates in low-accuracy mode but consumes less current.                                                   |                                                                                                                                                                                                                                                                                          |
|       | 1          | HIGH                                                                                                                                                                       | ACMP operates in high-accuracy mode but consumes more current.                                                  |                                                                                                                                                                                                                                                                                          |
| 16    | INPUTRANGE | 0x0                                                                                                                                                                        | RW                                                                                                              | <b>Input Range</b><br><br>Adjust performance of the comparator for a given input voltage range.                                                                                                                                                                                          |
|       | Value      | Mode                                                                                                                                                                       | Description                                                                                                     |                                                                                                                                                                                                                                                                                          |
|       | 0          | FULL                                                                                                                                                                       | Use this setting when the input to the comparator core can be from 0 to AVDD.                                   |                                                                                                                                                                                                                                                                                          |
|       | 1          | REDUCED                                                                                                                                                                    | It is recommended to use this setting when the input to the comparator core will always be less than AVDD-0.7V. |                                                                                                                                                                                                                                                                                          |
| 15:12 | Reserved   | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |                                                                                                                 |                                                                                                                                                                                                                                                                                          |
| 11:8  | HYST       | 0x0                                                                                                                                                                        | RW                                                                                                              | <b>Hysteresis mode</b><br><br>Set hysteresis mode and level.                                                                                                                                                                                                                             |
|       | Value      | Mode                                                                                                                                                                       | Description                                                                                                     |                                                                                                                                                                                                                                                                                          |
|       | 0          | DISABLED                                                                                                                                                                   | Hysteresis disabled                                                                                             |                                                                                                                                                                                                                                                                                          |
|       | 1          | SYM10MV                                                                                                                                                                    | 10mV symmetrical hysteresis                                                                                     |                                                                                                                                                                                                                                                                                          |
|       | 2          | SYM20MV                                                                                                                                                                    | 20mV symmetrical hysteresis                                                                                     |                                                                                                                                                                                                                                                                                          |
|       | 3          | SYM30MV                                                                                                                                                                    | 30mV symmetrical hysteresis                                                                                     |                                                                                                                                                                                                                                                                                          |
|       | 4          | POS10MV                                                                                                                                                                    | 10mV hysteresis on positive edge transitions                                                                    |                                                                                                                                                                                                                                                                                          |
|       | 5          | POS20MV                                                                                                                                                                    | 20mV hysteresis on positive edge transitions                                                                    |                                                                                                                                                                                                                                                                                          |

| Bit                                                                       | Name     | Reset                                                                                                                                                                      | Access | Description                                  |
|---------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------|
|                                                                           | 6        | POS30MV                                                                                                                                                                    |        | 30mV hysteresis on positive edge transitions |
|                                                                           | 8        | NEG10MV                                                                                                                                                                    |        | 10mV hysteresis on negative edge transitions |
|                                                                           | 9        | NEG20MV                                                                                                                                                                    |        | 20mV hysteresis on negative edge transitions |
|                                                                           | 10       | NEG30MV                                                                                                                                                                    |        | 30mV hysteresis on negative edge transitions |
| 7:3                                                                       | Reserved | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                              |
| 2:0                                                                       | BIAS     | 0x4                                                                                                                                                                        | RW     | <b>Bias Configuration</b>                    |
| These bits control the bias current level. See the datasheet for details. |          |                                                                                                                                                                            |        |                                              |

### 29.5.5 ACMP\_CTRL - Control Register

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |         |           |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|---------|-----------|
| 0x010  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |         |           |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | 0x0     | 0x0       |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | RW      | RW        |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | GPIOINV | NOTRDYVAL |

| Bit                                                                                      | Name      | Reset                                                                                                                                                                      | Access                                        | Description                          |
|------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------|
| 31:2                                                                                     | Reserved  | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |                                               |                                      |
| 1                                                                                        | GPIOINV   | 0x0                                                                                                                                                                        | RW                                            | <b>Comparator GPIO Output Invert</b> |
| Set this bit to 1 to invert the comparator alternate function output to GPIO.            |           |                                                                                                                                                                            |                                               |                                      |
|                                                                                          | Value     | Mode                                                                                                                                                                       | Description                                   |                                      |
|                                                                                          | 0         | NOTINV                                                                                                                                                                     | The comparator output to GPIO is not inverted |                                      |
|                                                                                          | 1         | INV                                                                                                                                                                        | The comparator output to GPIO is inverted     |                                      |
| 0                                                                                        | NOTRDYVAL | 0x0                                                                                                                                                                        | RW                                            | <b>Not Ready Value</b>               |
| The value of this bit is used as the comparator output when the comparator is not ready. |           |                                                                                                                                                                            |                                               |                                      |
|                                                                                          | Value     | Mode                                                                                                                                                                       | Description                                   |                                      |
|                                                                                          | 0         | LOW                                                                                                                                                                        | ACMP output is 0 when the ACMP is not ready.  |                                      |
|                                                                                          | 1         | HIGH                                                                                                                                                                       | ACMP output is 1 when the ACMP is not ready.  |                                      |

## 29.5.6 ACMP\_INPUTCTRL - Input Control Register

| Offset | Bit Position |    |          |    |    |    |    |    |    |    |    |    |         |    |    |    |    |    |    |        |    |    |   |   |   |   |   |        |   |   |   |   |
|--------|--------------|----|----------|----|----|----|----|----|----|----|----|----|---------|----|----|----|----|----|----|--------|----|----|---|---|---|---|---|--------|---|---|---|---|
| 0x014  | 31           | 30 | 29       | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19      | 18 | 17 | 16 | 15 | 14 | 13 | 12     | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4      | 3 | 2 | 1 | 0 |
| Reset  |              |    | 0x0      |    |    |    |    |    |    |    |    |    | 0x0     |    |    |    |    |    |    | 0x0    |    |    |   |   |   |   |   | 0x0    |   |   |   |   |
| Access |              |    | RW       |    |    |    |    |    |    |    |    |    | RW      |    |    |    |    |    |    | RW     |    |    |   |   |   |   |   | RW     |   |   |   |   |
| Name   |              |    | CSRESSEL |    |    |    |    |    |    |    |    |    | VREFDIV |    |    |    |    |    |    | NEGSEL |    |    |   |   |   |   |   | POSSEL |   |   |   |   |

| Bit   | Name     | Reset                                                                                                                                                                      | Access | Description                                                                                                              |
|-------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------|
| 31    | Reserved | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                                                                                                          |
| 30:28 | CSRESSEL | 0x0                                                                                                                                                                        | RW     | <b>Capacitive Sense Mode Internal Resistor</b><br>Deprecated capacitive sensing feature, not recommended for new designs |
|       | Value    | Mode                                                                                                                                                                       |        | Description                                                                                                              |
|       | 0        | RES0                                                                                                                                                                       |        | Internal capacitive sense resistor value 0                                                                               |
|       | 1        | RES1                                                                                                                                                                       |        | Internal capacitive sense resistor value 1                                                                               |
|       | 2        | RES2                                                                                                                                                                       |        | Internal capacitive sense resistor value 2                                                                               |
|       | 3        | RES3                                                                                                                                                                       |        | Internal capacitive sense resistor value 3                                                                               |
|       | 4        | RES4                                                                                                                                                                       |        | Internal capacitive sense resistor value 4                                                                               |
|       | 5        | RES5                                                                                                                                                                       |        | Internal capacitive sense resistor value 5                                                                               |
|       | 6        | RES6                                                                                                                                                                       |        | Internal capacitive sense resistor value 6                                                                               |
| 27:22 | Reserved | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                                                                                                          |
| 21:16 | VREFDIV  | 0x0                                                                                                                                                                        | RW     | <b>VREF division</b><br>Set division factor for VREFDIV. $VREFOUT = VREFIN * (VREFDIV / 63)$                             |
| 15:8  | NEGSEL   | 0x0                                                                                                                                                                        | RW     | <b>Negative Input Select</b><br>Select negative input.                                                                   |
|       | Value    | Mode                                                                                                                                                                       |        | Description                                                                                                              |
|       | 0        | VSS                                                                                                                                                                        |        | VSS                                                                                                                      |
|       | 16       | VREFDIVAVDD                                                                                                                                                                |        | Divided AVDD                                                                                                             |
|       | 17       | VREFDIVAVDDL                                                                                                                                                               |        | Low-Power Divided AVDD                                                                                                   |
|       | 18       | VREFDIV1V25                                                                                                                                                                |        | Divided 1V25 reference                                                                                                   |
|       | 19       | VREFDIV1V25LP                                                                                                                                                              |        | Low-power Divided 1V25 reference                                                                                         |
|       | 20       | VREFDIV2V5                                                                                                                                                                 |        | Divided 2V5 reference                                                                                                    |
|       | 21       | VREFDIV2V5LP                                                                                                                                                               |        | Low-power Divided 2V5 reference                                                                                          |
|       | 32       | VSENSE0DIV4                                                                                                                                                                |        | VSENSE0 divided by 4                                                                                                     |
|       | 33       | VSENSE0DIV4LP                                                                                                                                                              |        | Low-power VSENSE0 divided by 4                                                                                           |

| Bit | Name | Reset          | Access | Description                                                            |
|-----|------|----------------|--------|------------------------------------------------------------------------|
| 34  |      | VSENSE11DIV4   |        | VSENSE1 divided by 4                                                   |
| 35  |      | VSENSE11DIV4LP |        | Low-power VSENSE1 divided by 4                                         |
| 48  |      | CAPSENSE       |        | Deprecated capacitive sensing feature, not recommended for new designs |
| 64  |      | VDAC0OUT0      |        | VDAC0 channel 0 output                                                 |
| 66  |      | VDAC1OUT0      |        | VDAC1 channel 0 output                                                 |
| 128 |      | PA0            |        | Port A, Pin0                                                           |
| 129 |      | PA1            |        | Port A, Pin1                                                           |
| 130 |      | PA2            |        | Port A, Pin2                                                           |
| 131 |      | PA3            |        | Port A, Pin3                                                           |
| 132 |      | PA4            |        | Port A, Pin4                                                           |
| 133 |      | PA5            |        | Port A, Pin5                                                           |
| 134 |      | PA6            |        | Port A, Pin6                                                           |
| 135 |      | PA7            |        | Port A, Pin7                                                           |
| 136 |      | PA8            |        | Port A, Pin8                                                           |
| 137 |      | PA9            |        | Port A, Pin9                                                           |
| 138 |      | PA10           |        | Port A, Pin10                                                          |
| 139 |      | PA11           |        | Port A, Pin11                                                          |
| 140 |      | PA12           |        | Port A, Pin12                                                          |
| 141 |      | PA13           |        | Port A, Pin13                                                          |
| 142 |      | PA14           |        | Port A, Pin14                                                          |
| 143 |      | PA15           |        | Port A, Pin15                                                          |
| 144 |      | PB0            |        | Port B, Pin0                                                           |
| 145 |      | PB1            |        | Port B, Pin1                                                           |
| 146 |      | PB2            |        | Port B, Pin2                                                           |
| 147 |      | PB3            |        | Port B, Pin3                                                           |
| 148 |      | PB4            |        | Port B, Pin4                                                           |
| 149 |      | PB5            |        | Port B, Pin5                                                           |
| 150 |      | PB6            |        | Port B, Pin6                                                           |
| 151 |      | PB7            |        | Port B, Pin7                                                           |
| 152 |      | PB8            |        | Port B, Pin8                                                           |
| 153 |      | PB9            |        | Port B, Pin9                                                           |
| 154 |      | PB10           |        | Port B, Pin10                                                          |
| 155 |      | PB11           |        | Port B, Pin11                                                          |
| 156 |      | PB12           |        | Port B, Pin12                                                          |
| 157 |      | PB13           |        | Port B, Pin13                                                          |
| 158 |      | PB14           |        | Port B, Pin14                                                          |

| Bit | Name                   | Reset | Access      | Description                  |
|-----|------------------------|-------|-------------|------------------------------|
|     | 159                    | PB15  |             | Port B, Pin15                |
|     | 160                    | PC0   |             | Port C, Pin0                 |
|     | 161                    | PC1   |             | Port C, Pin1                 |
|     | 162                    | PC2   |             | Port C, Pin2                 |
|     | 163                    | PC3   |             | Port C, Pin3                 |
|     | 164                    | PC4   |             | Port C, Pin4                 |
|     | 165                    | PC5   |             | Port C, Pin5                 |
|     | 166                    | PC6   |             | Port C, Pin6                 |
|     | 167                    | PC7   |             | Port C, Pin7                 |
|     | 168                    | PC8   |             | Port C, Pin8                 |
|     | 169                    | PC9   |             | Port C, Pin9                 |
|     | 170                    | PC10  |             | Port C, Pin10                |
|     | 171                    | PC11  |             | Port C, Pin11                |
|     | 172                    | PC12  |             | Port C, Pin12                |
|     | 173                    | PC13  |             | Port C, Pin13                |
|     | 174                    | PC14  |             | Port C, Pin14                |
|     | 175                    | PC15  |             | Port C, Pin15                |
|     | 176                    | PD0   |             | Port D, Pin0                 |
|     | 177                    | PD1   |             | Port D, Pin1                 |
|     | 178                    | PD2   |             | Port D, Pin2                 |
|     | 179                    | PD3   |             | Port D, Pin3                 |
|     | 180                    | PD4   |             | Port D, Pin4                 |
|     | 181                    | PD5   |             | Port D, Pin5                 |
|     | 182                    | PD6   |             | Port D, Pin6                 |
|     | 183                    | PD7   |             | Port D, Pin7                 |
|     | 184                    | PD8   |             | Port D, Pin8                 |
|     | 185                    | PD9   |             | Port D, Pin9                 |
|     | 186                    | PD10  |             | Port D, Pin10                |
|     | 187                    | PD11  |             | Port D, Pin11                |
|     | 188                    | PD12  |             | Port D, Pin12                |
|     | 189                    | PD13  |             | Port D, Pin13                |
|     | 190                    | PD14  |             | Port D, Pin14                |
|     | 191                    | PD15  |             | Port D, Pin15                |
| 7:0 | POSSEL                 | 0x0   | RW          | <b>Positive Input Select</b> |
|     | Select positive input. |       |             |                              |
|     | Value                  | Mode  | Description |                              |

| Bit | Name | Reset          | Access | Description                      |
|-----|------|----------------|--------|----------------------------------|
| 0   |      | VSS            |        | VSS                              |
| 16  |      | VREFDIVAVDD    |        | Divided AVDD                     |
| 17  |      | VREFDIVAVDDL   |        | Low-Power Divided AVDD           |
| 18  |      | VREFDIV1V25    |        | Divided 1V25 reference           |
| 19  |      | VREFDIV1V25LP  |        | Low-power Divided 1V25 reference |
| 20  |      | VREFDIV2V5     |        | Divided 2V5 reference            |
| 21  |      | VREFDIV2V5LP   |        | Low-power Divided 2V5 reference  |
| 32  |      | VSENSE01DIV4   |        | VSENSE0 divided by 4             |
| 33  |      | VSENSE01DIV4LP |        | Low-power VSENSE0 divided by 4   |
| 34  |      | VSENSE11DIV4   |        | VSENSE1 divided by 4             |
| 35  |      | VSENSE11DIV4LP |        | Low-power VSENSE1 divided by 4   |
| 65  |      | VDAC0OUT1      |        | VDAC0 channel 1 output           |
| 67  |      | VDAC1OUT1      |        | VDAC1 channel 1 output           |
| 80  |      | EXTPA          |        | External interface, base is PA0. |
| 81  |      | EXTPB          |        | External interface, base is PB0. |
| 82  |      | EXTPC          |        | External interface, base is PC0. |
| 83  |      | EXTPD          |        | External interface, base is PD0. |
| 128 |      | PA0            |        | Port A, Pin0                     |
| 129 |      | PA1            |        | Port A, Pin1                     |
| 130 |      | PA2            |        | Port A, Pin2                     |
| 131 |      | PA3            |        | Port A, Pin3                     |
| 132 |      | PA4            |        | Port A, Pin4                     |
| 133 |      | PA5            |        | Port A, Pin5                     |
| 134 |      | PA6            |        | Port A, Pin6                     |
| 135 |      | PA7            |        | Port A, Pin7                     |
| 136 |      | PA8            |        | Port A, Pin8                     |
| 137 |      | PA9            |        | Port A, Pin9                     |
| 138 |      | PA10           |        | Port A, Pin10                    |
| 139 |      | PA11           |        | Port A, Pin11                    |
| 140 |      | PA12           |        | Port A, Pin12                    |
| 141 |      | PA13           |        | Port A, Pin13                    |
| 142 |      | PA14           |        | Port A, Pin14                    |
| 143 |      | PA15           |        | Port A, Pin15                    |
| 144 |      | PB0            |        | Port B, Pin0                     |
| 145 |      | PB1            |        | Port B, Pin1                     |
| 146 |      | PB2            |        | Port B, Pin2                     |
| 147 |      | PB3            |        | Port B, Pin3                     |

| Bit | Name | Reset | Access | Description   |
|-----|------|-------|--------|---------------|
| 148 |      | PB4   |        | Port B, Pin4  |
| 149 |      | PB5   |        | Port B, Pin5  |
| 150 |      | PB6   |        | Port B, Pin6  |
| 151 |      | PB7   |        | Port B, Pin7  |
| 152 |      | PB8   |        | Port B, Pin8  |
| 153 |      | PB9   |        | Port B, Pin9  |
| 154 |      | PB10  |        | Port B, Pin10 |
| 155 |      | PB11  |        | Port B, Pin11 |
| 156 |      | PB12  |        | Port B, Pin12 |
| 157 |      | PB13  |        | Port B, Pin13 |
| 158 |      | PB14  |        | Port B, Pin14 |
| 159 |      | PB15  |        | Port B, Pin15 |
| 160 |      | PC0   |        | Port C, Pin0  |
| 161 |      | PC1   |        | Port C, Pin1  |
| 162 |      | PC2   |        | Port C, Pin2  |
| 163 |      | PC3   |        | Port C, Pin3  |
| 164 |      | PC4   |        | Port C, Pin4  |
| 165 |      | PC5   |        | Port C, Pin5  |
| 166 |      | PC6   |        | Port C, Pin6  |
| 167 |      | PC7   |        | Port C, Pin7  |
| 168 |      | PC8   |        | Port C, Pin8  |
| 169 |      | PC9   |        | Port C, Pin9  |
| 170 |      | PC10  |        | Port C, Pin10 |
| 171 |      | PC11  |        | Port C, Pin11 |
| 172 |      | PC12  |        | Port C, Pin12 |
| 173 |      | PC13  |        | Port C, Pin13 |
| 174 |      | PC14  |        | Port C, Pin14 |
| 175 |      | PC15  |        | Port C, Pin15 |
| 176 |      | PD0   |        | Port D, Pin0  |
| 177 |      | PD1   |        | Port D, Pin1  |
| 178 |      | PD2   |        | Port D, Pin2  |
| 179 |      | PD3   |        | Port D, Pin3  |
| 180 |      | PD4   |        | Port D, Pin4  |
| 181 |      | PD5   |        | Port D, Pin5  |
| 182 |      | PD6   |        | Port D, Pin6  |
| 183 |      | PD7   |        | Port D, Pin7  |
| 184 |      | PD8   |        | Port D, Pin8  |

| Bit | Name | Reset | Access | Description   |
|-----|------|-------|--------|---------------|
|     | 185  | PD9   |        | Port D, Pin9  |
|     | 186  | PD10  |        | Port D, Pin10 |
|     | 187  | PD11  |        | Port D, Pin11 |
|     | 188  | PD12  |        | Port D, Pin12 |
|     | 189  | PD13  |        | Port D, Pin13 |
|     | 190  | PD14  |        | Port D, Pin14 |
|     | 191  | PD15  |        | Port D, Pin15 |

### 29.5.7 ACMP\_STATUS - Status Register

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |              |               |         |   |         |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|--------------|---------------|---------|---|---------|
| 0x018  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3            | 2             | 1       | 0 |         |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | 0x0          | 0x0           | 0x0     |   | 0x0     |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | R            | R             | R       |   | R       |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | PORTALLOCERR | INPUTCONFLICT | ACMPRDY |   | ACMPOUT |

| Bit  | Name          | Reset                                                                                                                                                                      | Access | Description                                                                                                                                                                             |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:5 | Reserved      | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                                                                                                                                                                         |
| 4    | PORTALLOCERR  | 0x0                                                                                                                                                                        | R      | <b>Port allocation error</b><br><br>The port selected by INPUTCTRL_POSSEL or INPUTCTRL_NEGSEL is not allocated to this ACMP. Port allocation needs to be configured in the GPIO module. |
| 3    | INPUTCONFLICT | 0x0                                                                                                                                                                        | R      | <b>INPUT conflict</b><br><br>INPUTCTRL_POSSEL and INPUTCTRL_NEGSEL is configured illegally.                                                                                             |
| 2    | ACMPRDY       | 0x0                                                                                                                                                                        | R      | <b>Analog Comparator Ready</b><br><br>Analog comparator ready status.                                                                                                                   |
| 1    | Reserved      | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                                                                                                                                                                         |
| 0    | ACMPOUT       | 0x0                                                                                                                                                                        | R      | <b>Analog Comparator Output</b><br><br>Analog comparator output value.                                                                                                                  |



### 29.5.8 ACMP\_IF - Interrupt Flag Register

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |             |               |         |      |      |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|-------------|---------------|---------|------|------|
| 0x01C  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3           | 2             | 1       | 0    |      |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | 0x0         | 0x0           | 0x0     | 0x0  | 0x0  |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | RW          | RW            | RW      | RW   | RW   |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | PORTALOCERR | INPUTCONFLICT | ACMPRDY | FALL | RISE |

| Bit  | Name          | Reset                                                                                                                                                                      | Access | Description                                                                                                                                                                                                                                                                                            |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:5 | Reserved      | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                                                                                                                                                                                                                                                                                        |
| 4    | PORTALLOCERR  | 0x0                                                                                                                                                                        | RW     | <b>Port allocation error</b><br><br>The port selected by INPUTCTRL_POSSEL or INPUTCTRL_NEGSEL is not allocated to this ACMP. Port allocation needs to be configured in the GPIO module.                                                                                                                |
| 3    | INPUTCONFLICT | 0x0                                                                                                                                                                        | RW     | <b>Input conflict</b><br><br>INPUTCTRL_POSSEL and INPUTCTRL_NEGSEL is configured illegally.                                                                                                                                                                                                            |
| 2    | ACMPRDY       | 0x0                                                                                                                                                                        | RW     | <b>ACMP ready Interrupt flag</b><br><br>Indicates that the analog comparator is ready and references have settled. Note at lower bias settings (BIAS<=3), the ACMPRDY bit may not be reliable and additional software wait may be required. Refer to "Warmup Time" section for additional information. |
| 1    | FALL          | 0x0                                                                                                                                                                        | RW     | <b>Falling Edge Triggered Interrupt Flag</b><br><br>Indicates that there has been a falling edge on the analog comparator output.                                                                                                                                                                      |
| 0    | RISE          | 0x0                                                                                                                                                                        | RW     | <b>Rising Edge Triggered Interrupt Flag</b><br><br>Indicates that there has been a rising edge on the analog comparator output.                                                                                                                                                                        |

### 29.5.9 ACMP\_IEN - Interrupt Enable Register

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |              |               |         |      |      |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|--------------|---------------|---------|------|------|
| 0x020  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3            | 2             | 1       | 0    |      |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | 0x0          | 0x0           | 0x0     | 0x0  | 0x0  |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | RW           | RW            | RW      | RW   | RW   |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   | PORTALLOCERR | INPUTCONFLICT | ACMPRDY | FALL | RISE |

| Bit  | Name          | Reset                                                                                                                                                                      | Access | Description                            |
|------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------|
| 31:5 | Reserved      | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                                        |
| 4    | PORTALLOCERR  | 0x0                                                                                                                                                                        | RW     | Port allocation error interrupt enable |
| 3    | INPUTCONFLICT | 0x0                                                                                                                                                                        | RW     | Input conflict interrupt enable        |
| 2    | ACMPRDY       | 0x0                                                                                                                                                                        | RW     | ACMP ready interrupt enable            |
| 1    | FALL          | 0x0                                                                                                                                                                        | RW     | Falling edge interrupt enable          |
| 0    | RISE          | 0x0                                                                                                                                                                        | RW     | Rising edge interrupt enable           |

### 29.5.10 ACMP\_SYNCBUSY - Syncbusy

| Offset | Bit Position |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   |           |
|--------|--------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|---|-----------|
| 0x024  | 31           | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |           |
| Reset  |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | 0x0       |
| Access |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | R         |
| Name   |              |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |   | INPUTCTRL |

| Bit  | Name                                 | Reset                                                                                                                                                                      | Access | Description            |
|------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------|
| 31:1 | Reserved                             | To ensure compatibility with future devices, always write Reserved bits to their reset value, unless otherwise stated. More information in <a href="#">1.2 Conventions</a> |        |                        |
| 0    | INPUTCTRL                            | 0x0                                                                                                                                                                        | R      | Syncbusy for INPUTCTRL |
|      | Synchronization of INPUTCTRL ongoing |                                                                                                                                                                            |        |                        |